Supplementary File 1 — GC–MS Compounds List

Sr. No.	RT	Compound Name	SMILES	Molecular Formula	Molecular Weight	M/Z	Area %
1	15.614	1-Butene, 1-(methylthio)-, (Z)-	CC/C=C\SC	C5H10S	102.19 g/mol	87	14.12
2	18.066	4-(2-Methyl-1H-imidazol-1-yl)aniline	Cc1nccn1-c1ccc(N)cc1	C10H11N3	173.22 g/mol	173.1	0.92
3	18.465	Phytol	C[C@@H](CCC[C@@H](C)CCC/C(=C/CO)/C)CCCC(C)C	C20H40O	296.53 g/mol	71	0.76
4	21.069	5-Cyano-4-oxo-6-phenyl-2-thioxohexahydropyrimidine	O=C1NC(=S)NC(c2ccccc2)C1C#N	C11H9N3OS	231.27 g/mol	231.2	2.02
5	21.989	Bis(2-ethylhexyl) phthalate	CCCCC(CC)COC(=O)c1ccccc1C(=O)OCC(CC)CCCC	C24H38O4	390.56 g/mol	149	10.39
6	22.216	3-Butenoic acid, 4-phenyl-, butyl ester	CCCCOC(=O)C/C=C/C1=CC=CC=C1	C14H18O2	218.30 g/mol	117.1	11.82
7	22.344	2-[(2-Furanylmethyl)amino]benzoic acid	O=C(O)c1ccccc1NCc1ccco1	C12H11NO3	217.22 g/mol	81	11.66
8	22.485	Tricyclo[3.2.1.0(2,4)]octane-3-carboxylic acid (furan-2-ylmethyl)-amide	O=C(NCc1ccco1)C1C2CC3CC(C2)C13	C14H17NO2	231.29 g/mol	162.1	5.77
9	22.998	8-Phosphadispiro[2.1.2.2]nonane, 8-phenyl-	C1=CC=C(C=C1)C2C(C(=O)NC(=S)N2)C#N	C14H17P	216.26 g/mol	216.1	33.82
10	23.329	6-Methoxy-2-[2-[imidazol-1-yl]ethoxy]-8-nitroquinoline	COc1cc([N+](=O)[O-])c2nc(OCCN3C=CN=C3)ccc2c1	C15H14N4O4	314.30 g/mol	95	2.52
11	23.464	16-Heptadecyn-4-one, 1,2-dihydroxy-	C#CCCCCCCCCCCCC(=O)CC(O)CO	C17H28O3	282.42 g/mol	81	1.14
12	27.75	Vitamin E (?-Tocopherol)	Cc1c(C)c2c(c(C)c1O)CC[C@](C)(CCC[C@H](C)CCC[C@H](C)CCCC(C)C)O2	C29H50O2	430.71 g/mol	165.1	2.69

GC–MS Instrument Control and Acquisition Parameters

INSTRUMENT CONTROL PARAMETERS: GCMS-5977B AGILENT TECHNOLOGIES USA.

D:\MassHunter\GCMS\1\methods\CRL Method .M

Control Information

Sample Inlet	: GC
Injection Source	: GC ALS
Injection Location:	Front
Mass Spectrometer	: Enabled

No Sample Prep method has been assigned to this method.

GC

GC Summary

Run Time	36 min
Post Run Time	0 min

Oven

Temperature	
Setpoint	On
(Initial)	70 °C
Hold Time	3 min
Post Run	50 °C
Program	
#1 Rate	10 °C/min
#1 Value	270 °C
#1 Hold Time	13 min

SOLVENT DELAY	2 mint
Max Temperature	300 °C
Maximum Temperature Override	Enabled
Slow Fan	Disabled

ALS

Front Injector	
Syringe Size	10 µL
Injection Volume	1 µL
Solvent A Washes (PreInj)	6
Solvent A Washes (PostInj)	6
Solvent A Volume	8 µL
Solvent B Washes (PreInj)	0
Solvent B Washes (PostInj)	0
Solvent B Volume	8 µL
Sample Washes	4
Sample Wash Volume	8 µL
Sample Pumps	4
Dwell Time (PreInj)	0 min
Dwell Time (PostInj)	0 min
Solvent Wash Draw Speed	300 µL/min
Solvent Wash Dispense Speed	3000 µL/min
Sample Wash Draw Speed	300 µL/min
Sample Wash Dispense Speed	3000 µL/min
Injection Dispense Speed	6000 µL/min
Viscosity Delay	0 sec
Sample Depth	1 mm
Injection Type	Standard
L1 Airgap	0.2 µL
Solvent Wash Mode	A, B

Sample Overlap	
Mode	Sample overlap is not enabled

ALS Errors	Pause for user interaction
Front SS Inlet He	

Mode	Split
Heater	On 250 °C
Pressure	On 8.8085 psi
Total Flow	On 24 mL/min
Septum Purge Flow	On 3 mL/min
Gas Saver	Off
Split Ratio	20 :1
Split Flow	20 mL/min

Thermal Aux 2 (MSD Transfer Line)

Temperature	
Setpoint	On
(Initial)	280 °C
Post Run	0 °C

Column

Column #1	
Flow	
Setpoint	On
(Initial)	1 mL/min
Post Run	1 mL/min

Column lock	Unlocked
In	Front SS Inlet He
Out	MSD
(Initial)	70 °C
Pressure	8.8085 psi
Flow	1 mL/min
Average Velocity	36.796 cm/sec
Holdup Time	1.3588 min

Column Outlet Pressure	0 psi

Signals

Signal #1: Test Plot	
Description	Test Plot
Details	
Save	Off
Data Rate	50 Hz
Dual Injection Assignment	Front Sample
Signal #2: Test Plot	
Description	Test Plot
Details	
Save	Off
Data Rate	50 Hz
Dual Injection Assignment	Back Sample
Signal #3: Test Plot	
Description	Test Plot
Details	
Save	Off
Data Rate	50 Hz
Dual Injection Assignment	Back Sample
Signal #4: Test Plot	
Description	Test Plot
Details	
Save	Off
Data Rate	50 Hz
Dual Injection Assignment	Back Sample

Column	DB-5
Length	-60 °C–325 °C (325 °C): 30 m x 250 µm x 0.25 µm
GAS USED	HELIUM

MS Information

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General Information

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Acquisition Mode	: Scan

```
Solvent Delay (minutes)  : 2
Tune file                : D:\MassHunter\GCMS\1\5977\ATUNE.U
EM Setting mode Gain     : 1.000000

Normal or Fast Scanning  : Normal Scanning
Trace Ion Detection      : Off
Run Time (if MS only)    : 650 minutes

[Scan Parameters]
Start Time               : 2
Low Mass                 : 30
High Mass                : 650
Threshold                : 150
A/D Samples:             : 4


[MSZones]

MS Source                   : 230 C   maximum 250 C
MS Quad                     : 150 C   maximum 200 C

Timed Events
----- ------
Number Events= 0



                   END OF MS ACQUISTION PARAMETERS
                      TUNE PARAMETERS for SN: US2048N001
                      ----------------------------------

 Trace Ion Detection is OFF.

    34.593   :   EMISSION
    70.007   :   ENERGY
    34.899   :   REPELLER
    90.331   :   IONFOCUS
    12.582   :   ENTRANCE_LENS
  1200.696   :   EMVOLTS
                            1082.8  : Actual EMV
                             1.00   : GAIN FACTOR
  2526.000   :   AMUGAIN
   141.375   :   AMUOFFSET
     1.000   :   FILAMENT
     1.000   :   DCPOLARITY
    12.689   :   ENTLENSOFFSET
     0.000   :   Ion_Body
     0.000   :   EXTLENS
    64.000   :  MASSGAIN
   -24.000   :  MASSOFFSET

                      END OF TUNE PARAMETERS
                      ----------------------



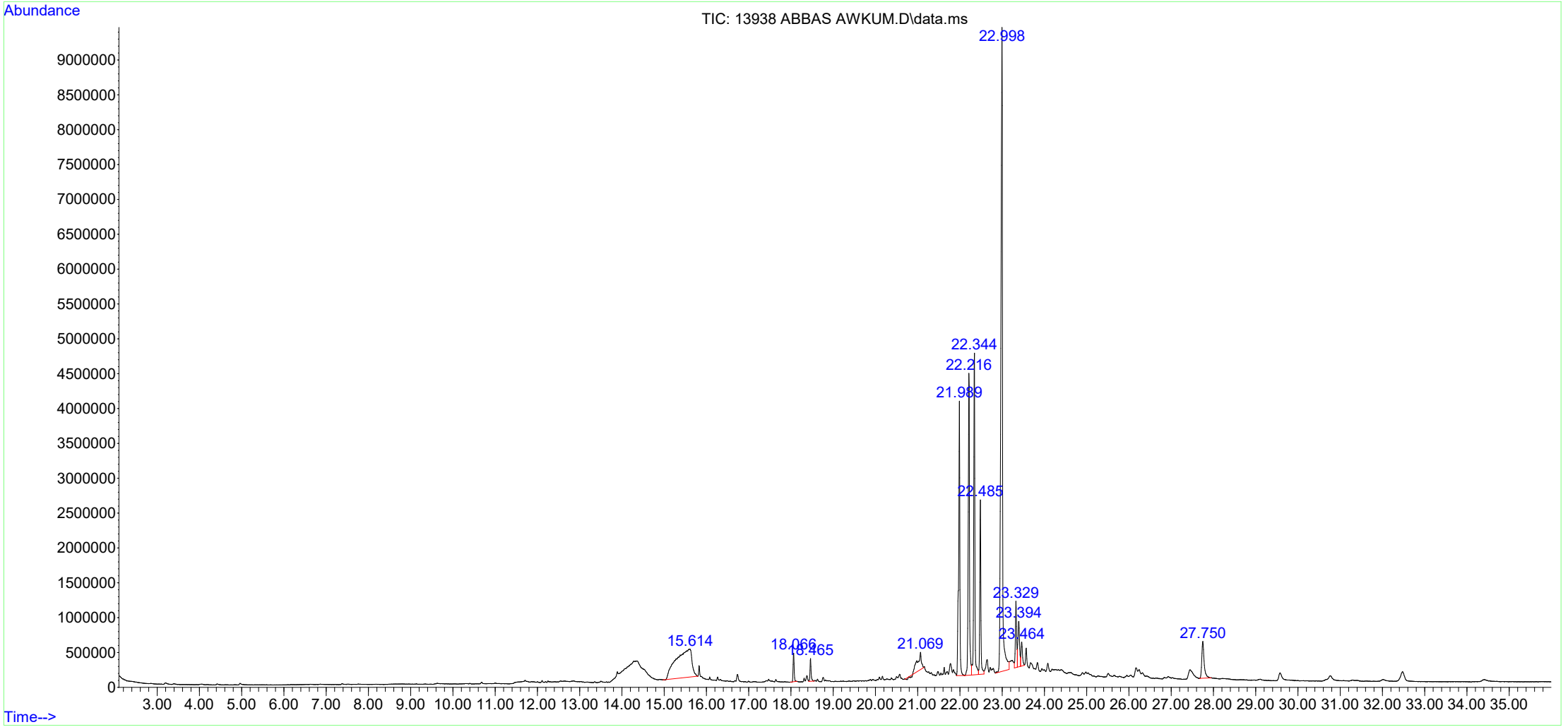
                     END OF INSTRUMENT CONTROL PARAMETERS
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Library Search Report

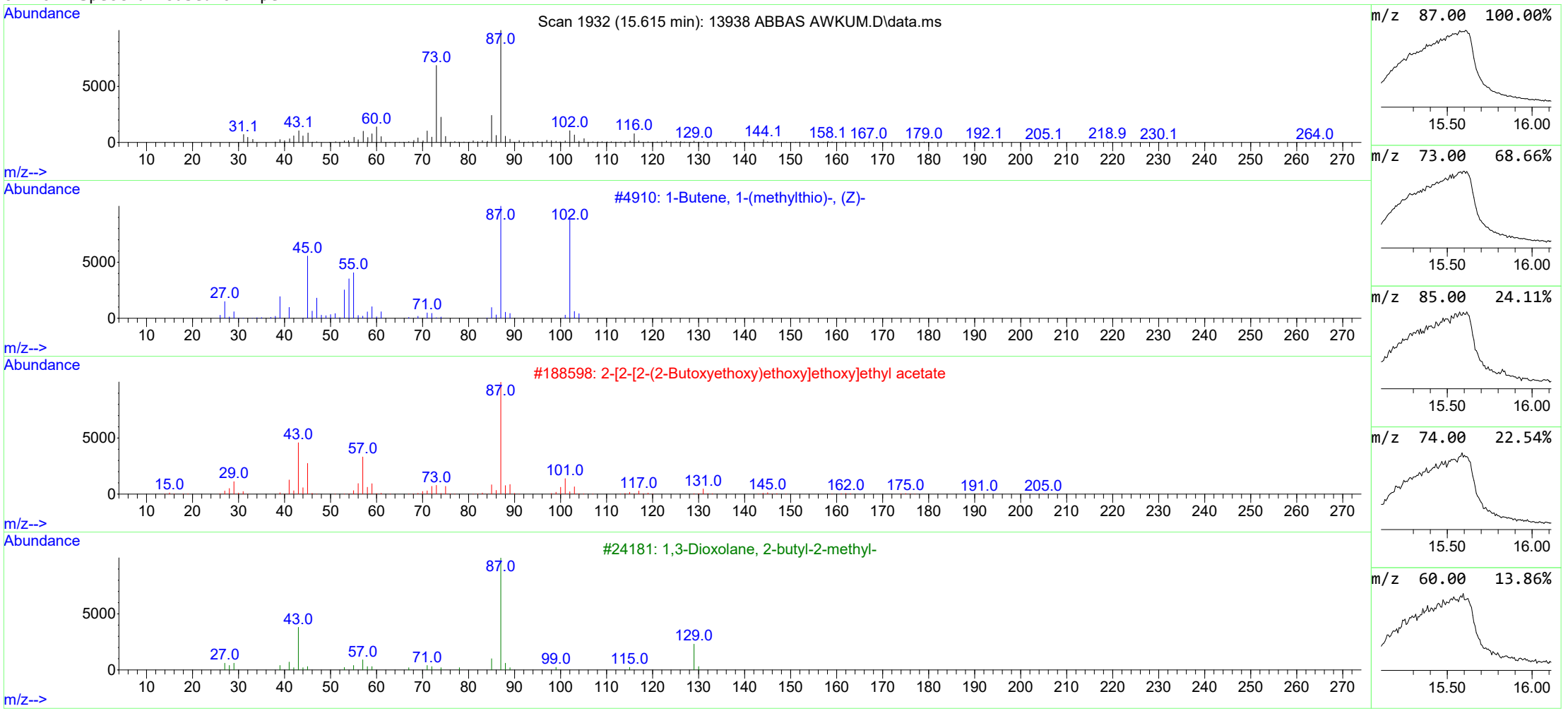
Data Path : D:\GCMS RESULTS DATA\
Data File : 13938 ABBAS AWKUM.D
Acq On : 08 Sep 2026 00:12
Operator : CRL
Sample : ABBAS
Misc :
ALS Vial : 12 Sample Multiplier: 1

Search Libraries: C:\MassHunter\LIBRARY\NIST20.L Minimum Quality: 0

Unknown Spectrum: Apex
Integration Events: ChemStation Integrator - autoint1.e



Unknown Spectrum based on Apex

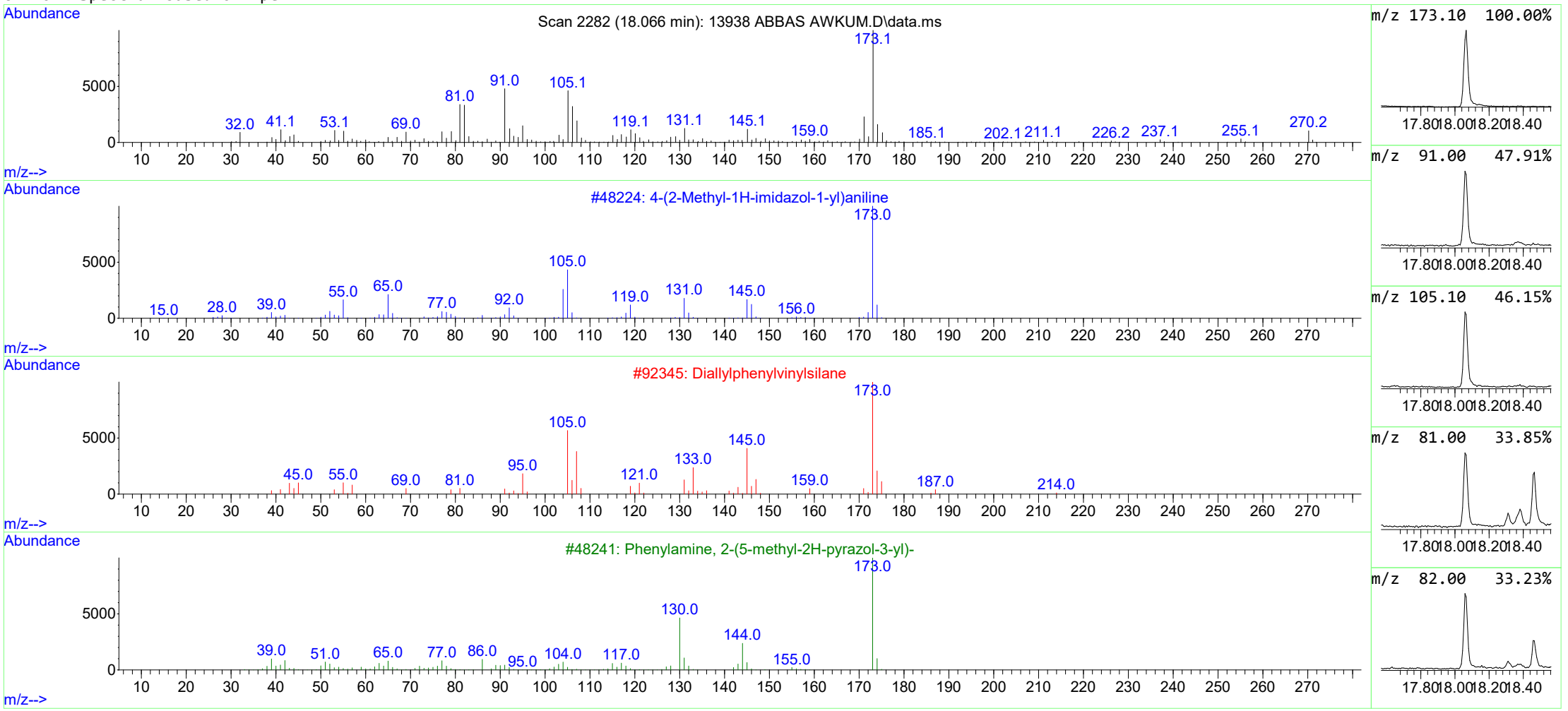


Data File: D:\GCMS RESULTS DATA\13938 ABBAS AWKUM.D
Sample : ABBAS

Peak Number: 1 at 15.614 min Area: 106588174 Area % 14.12

The 3 best hits from each library.	Ref\#	CAS\#	Qual
C:\MassHunter\LIBRARY\NIST20.L			
1 1-Butene, 1-(methylthio)-, (Z)-	4910	017414-15-2	47
2 2-[2-[2-(2-Butoxyethoxy)ethoxy]e...	188598	1000351-93-9	43
3 1,3-Dioxolane, 2-butyl-2-methyl-	24181	014447-27-9	43

Unknown Spectrum based on Apex



Data File: D:\GCMS RESULTS DATA\13938 ABBAS AWKUM.D
Sample : ABBAS

Peak Number: 2 at 18.066 min Area: 6965043 Area % 0.92

The 3 best hits from each library.	Ref\#	CAS\#	Qual
C:\MassHunter\LIBRARY\NIST20.L			
1 4-(2-Methyl-1H-imidazol-1-yl)ani...	48224	1000484-51-3	58
2 Diallylphenylvinylsilane	92345	119901-89-2	50
3 Phenylamine, 2-(5-methyl-2H-pyra...	48241	1000309-95-3	43

Unknown Spectrum based on Apex

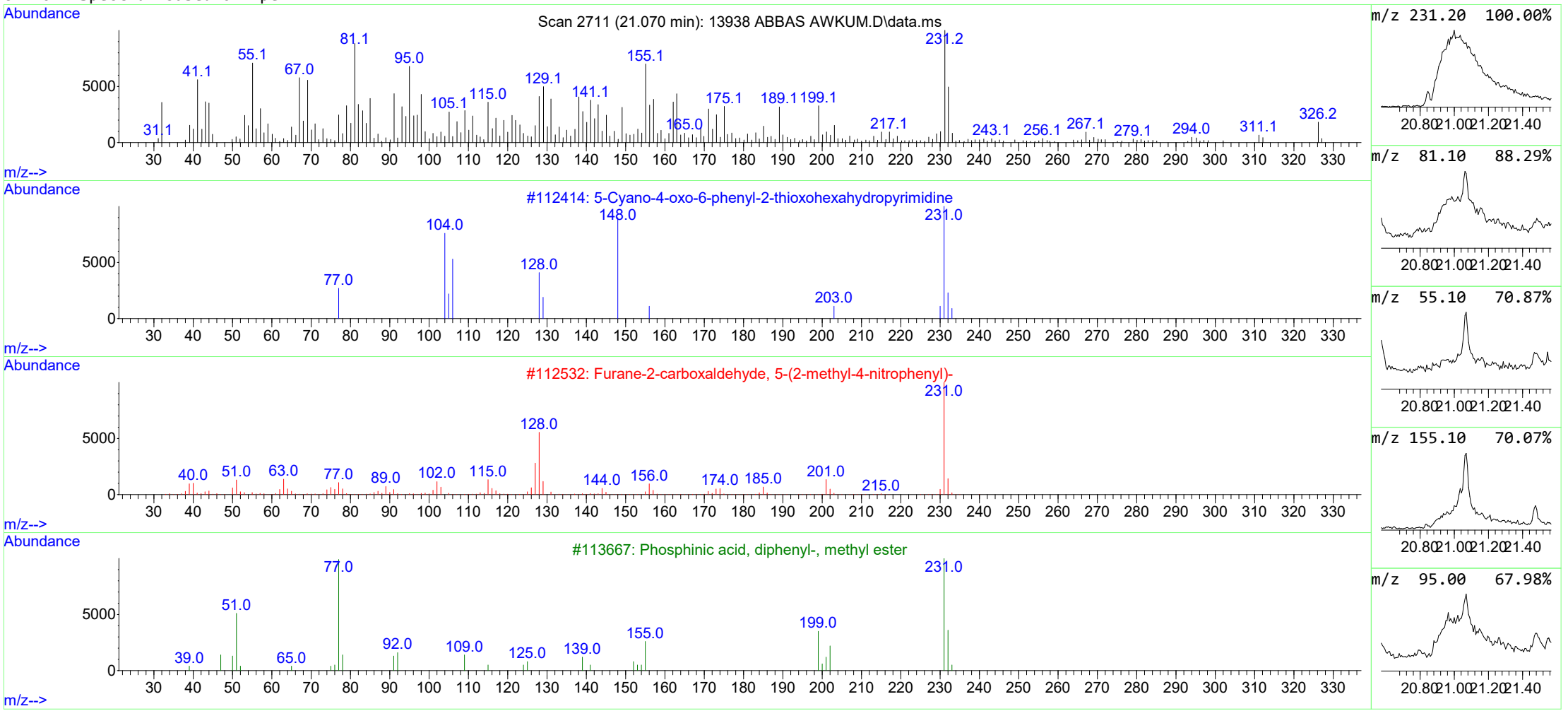


Data File: D:\GCMS RESULTS DATA\13938 ABBAS AWKUM.D
Sample : ABBAS

Peak Number: 3 at 18.465 min Area: 5751144 Area % 0.76

The 3 best hits from each library.	Ref\#	CAS\#	Qual
C:\MassHunter\LIBRARY\NIST20.L			
1 Phytol	194548	000150-86-7	99
2 Phytol	194550	000150-86-7	91
3 Phytol	194552	000150-86-7	91

Unknown Spectrum based on Apex

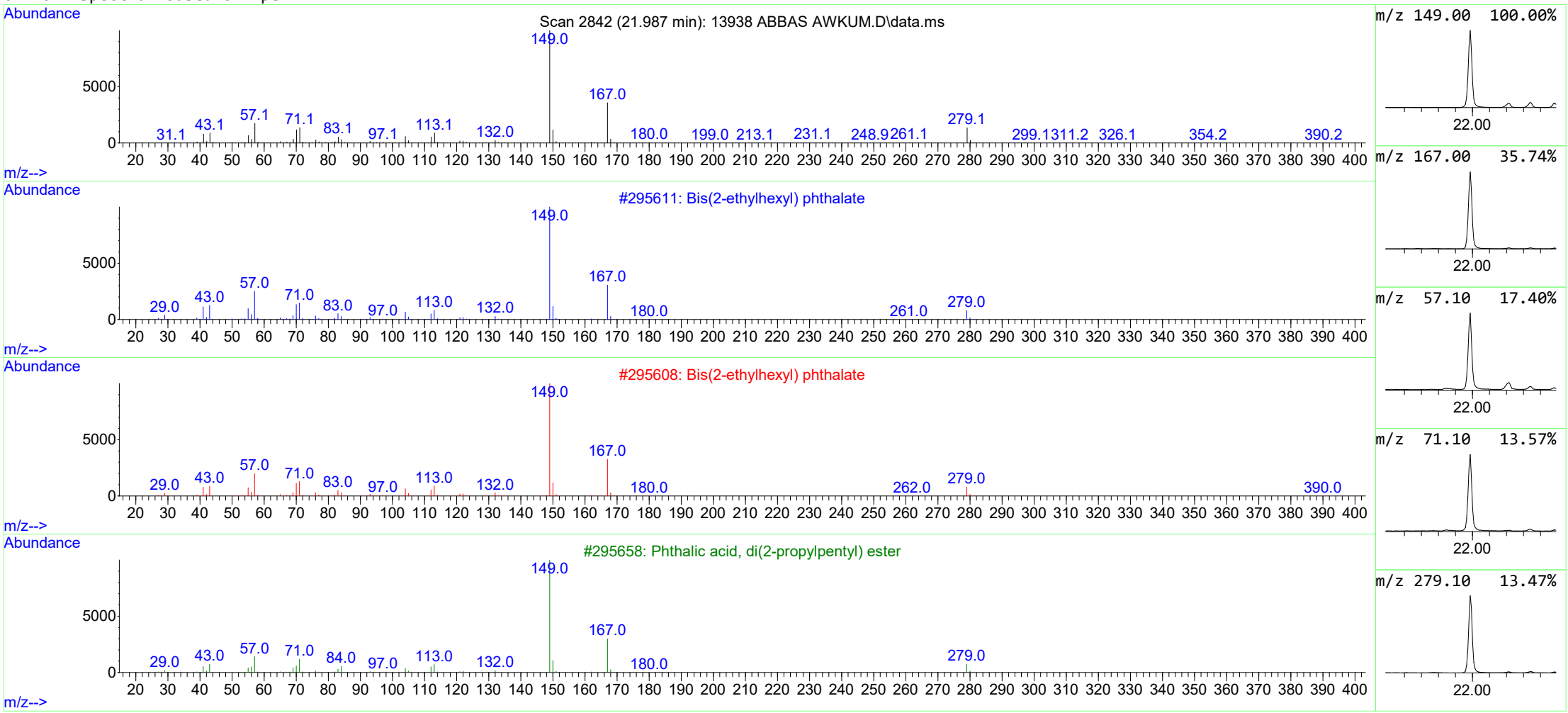


Data File: D:\GCMS RESULTS DATA\13938 ABBAS AWKUM.D
Sample : ABBAS

Peak Number: 4 at 21.069 min Area: 15242680 Area % 2.02

The 3 best hits from each library.	Ref\#	CAS\#	Qual
C:\MassHunter\LIBRARY\NIST20.L			
1 5-Cyano-4-oxo-6-phenyl-2-thioxoh...	112414	081585-29-7	25
2 Furane-2-carboxaldehyde, 5-(2-me...	112532	329222-70-0	22
3 Phosphinic acid, diphenyl-, meth...	113667	001706-90-7	18

Unknown Spectrum based on Apex

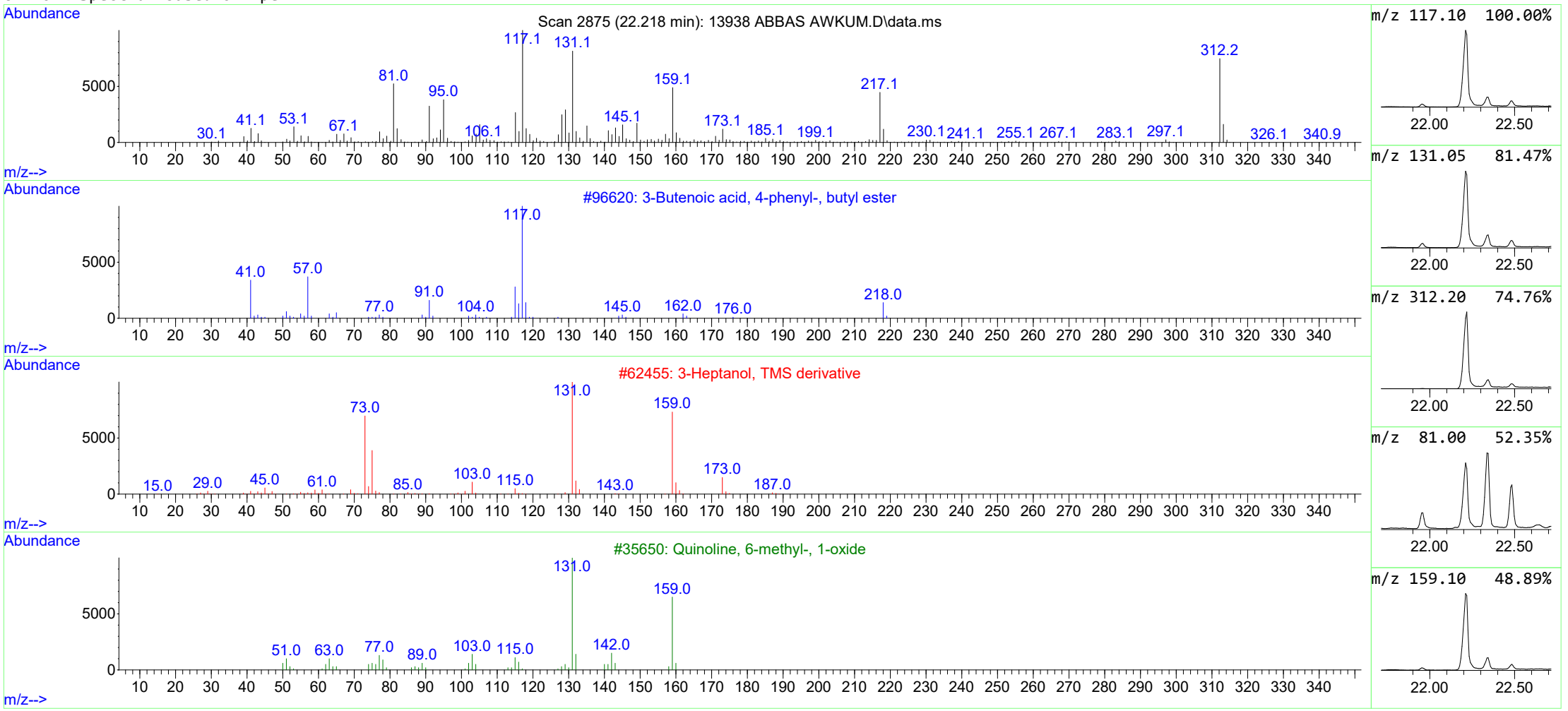


Data File: D:\GCMS RESULTS DATA\13938 ABBAS AWKUM.D
Sample : ABBAS

Peak Number: 5 at 21.989 min Area: 78443017 Area % 10.39

The 3 best hits from each library.	Ref\#	CAS\#	Qual
C:\MassHunter\LIBRARY\NIST20.L			
1 Bis(2-ethylhexyl) phthalate	295611	000117-81-7	91
2 Bis(2-ethylhexyl) phthalate	295608	000117-81-7	91
3 Phthalic acid, di(2-propylpentyl...	295658	1000377-93-5	91

Unknown Spectrum based on Apex

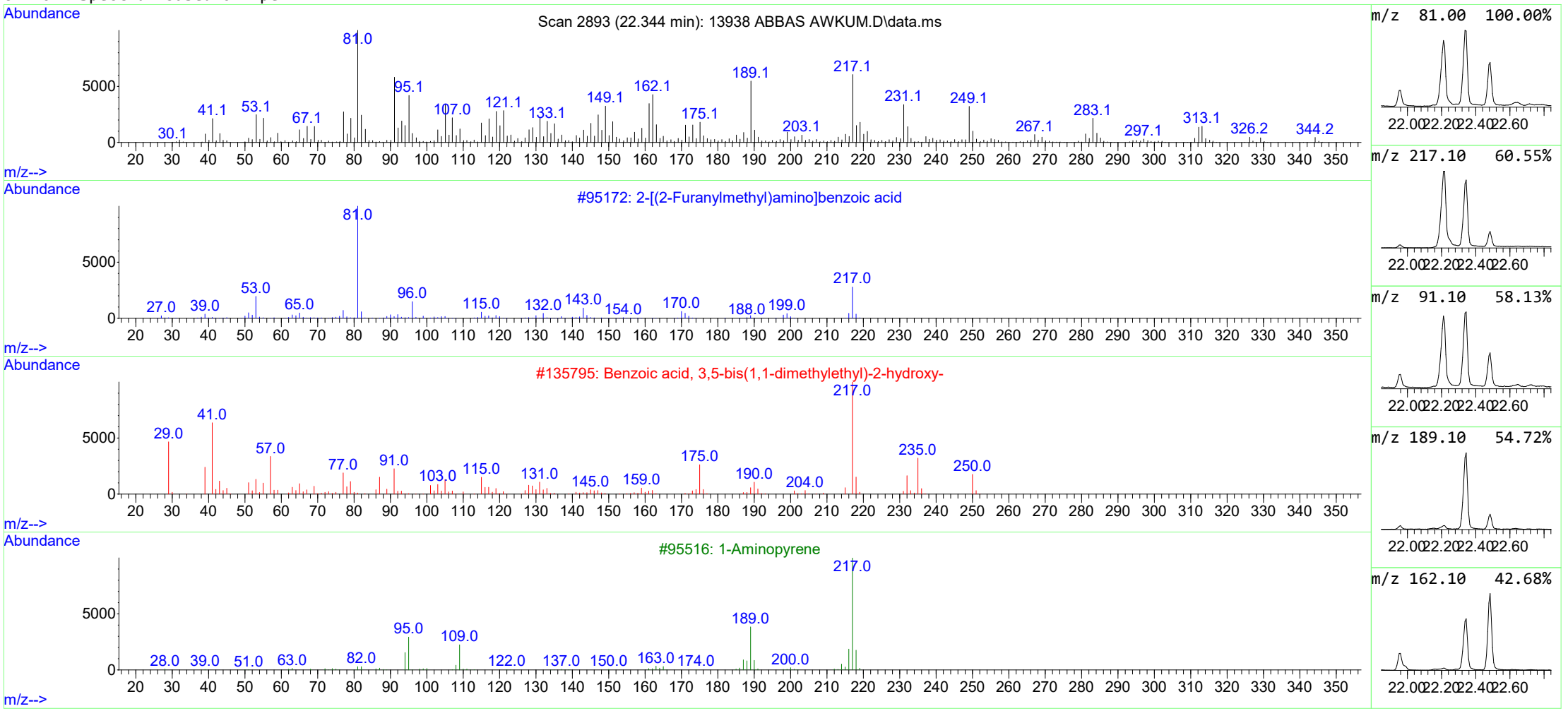


Data File: D:\GCMS RESULTS DATA\13938 ABBAS AWKUM.D
Sample : ABBAS

Peak Number: 6 at 22.216 min Area: 89233175 Area % 11.82

The 3 best hits from each library.	Ref\#	CAS\#	Qual
C:\MassHunter\LIBRARY\NIST20.L			
1 3-Butenoic acid, 4-phenyl-, buty...	96620	054966-44-8	25
2 3-Heptanol, TMS derivative	62455	018132-91-7	25
3 Quinoline, 6-methyl-, 1-oxide	35650	004053-42-3	14

Unknown Spectrum based on Apex

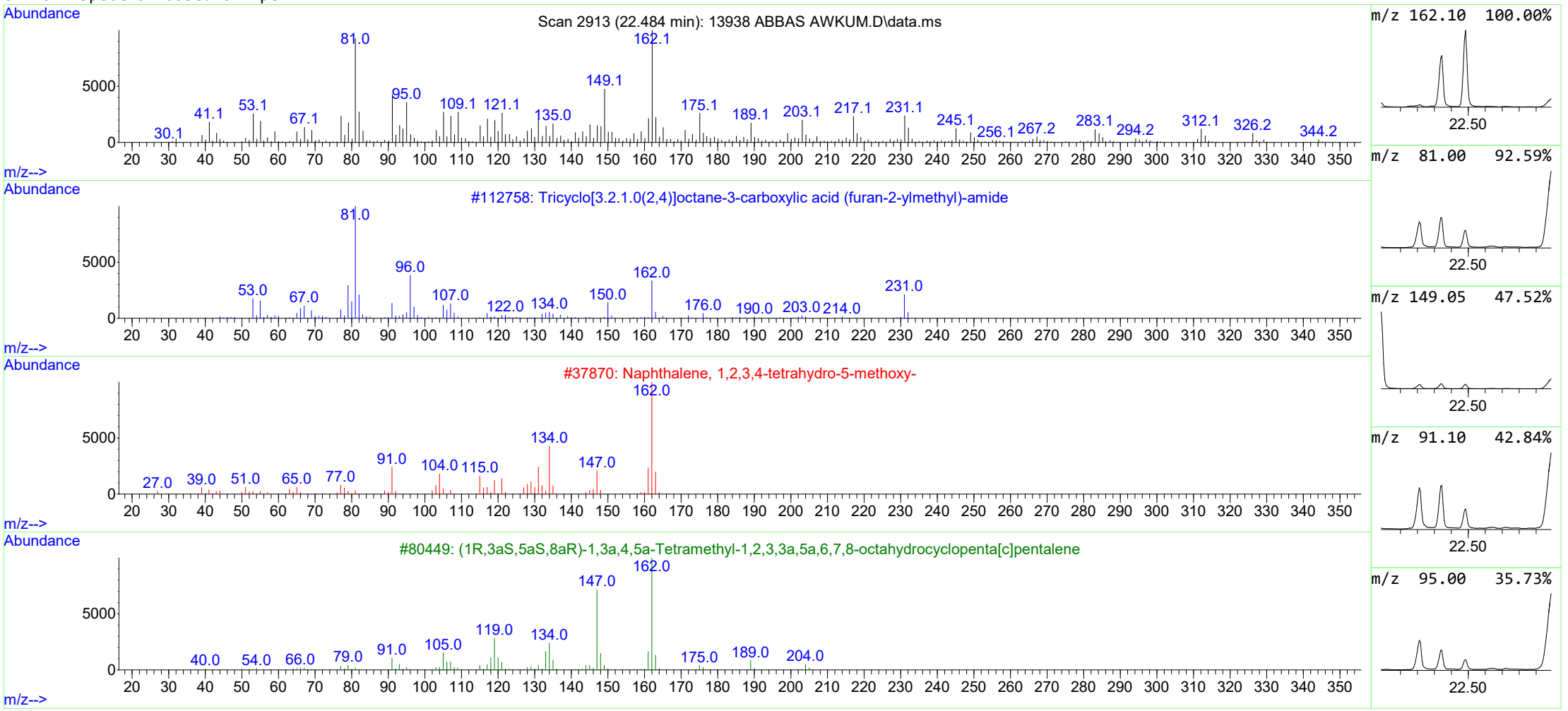


Data File: D:\GCMS RESULTS DATA\13938 ABBAS AWKUM.D
Sample : ABBAS

Peak Number: 7 at 22.344 min Area: 88035449 Area % 11.66

The 3 best hits from each library.	Ref\#	CAS\#	Qual
C:\MassHunter\LIBRARY\NIST20.L			
1 2-[(2-Furanylmethyl)amino]benzoi...	95172	501661-50-3	30
2 Benzoic acid, 3,5-bis(1,1-dimeth...	135795	019715-19-6	25
3 1-Aminopyrene	95516	001606-67-3	25

Unknown Spectrum based on Apex

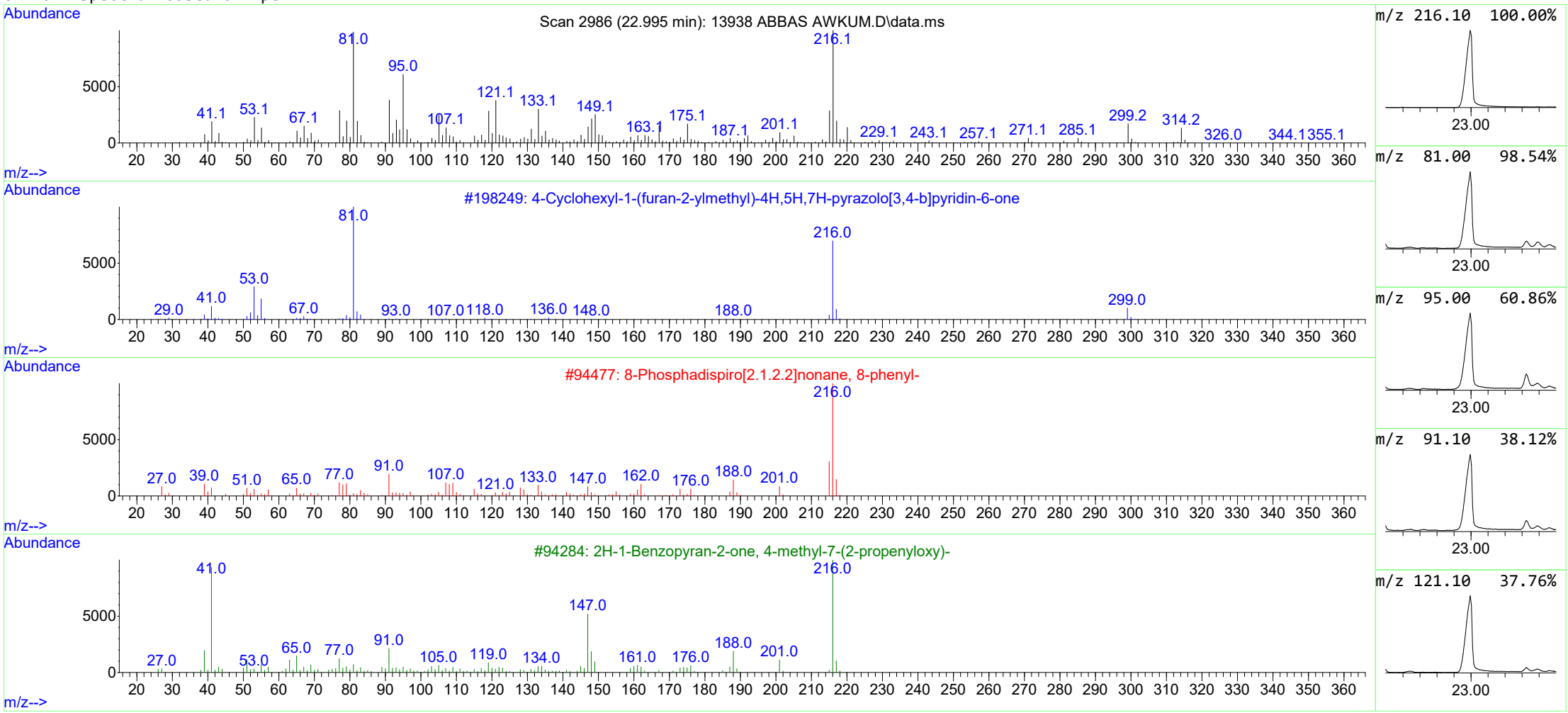


Data File: D:\GCMS RESULTS DATA\13938 ABBAS AWKUM.D
Sample : ABBAS

Peak Number: 8 at 22.485 min Area: 43531743 Area % 5.77

The 3 best hits from each library.	Ref\#	CAS\#	Qual
C:\MassHunter\LIBRARY\NIST20.L			
1 Tricyclo[3.2.1.0(2,4)]octane-3-c...	112758	1000297-39-2	30
2 Naphthalene, 1,2,3,4-tetrahydro-...	37870	001008-19-1	25
3 (1R,3aS,5aS,8aR)-1,3a,4,5a-Tetra...	80449	065372-78-3	20

Unknown Spectrum based on Apex

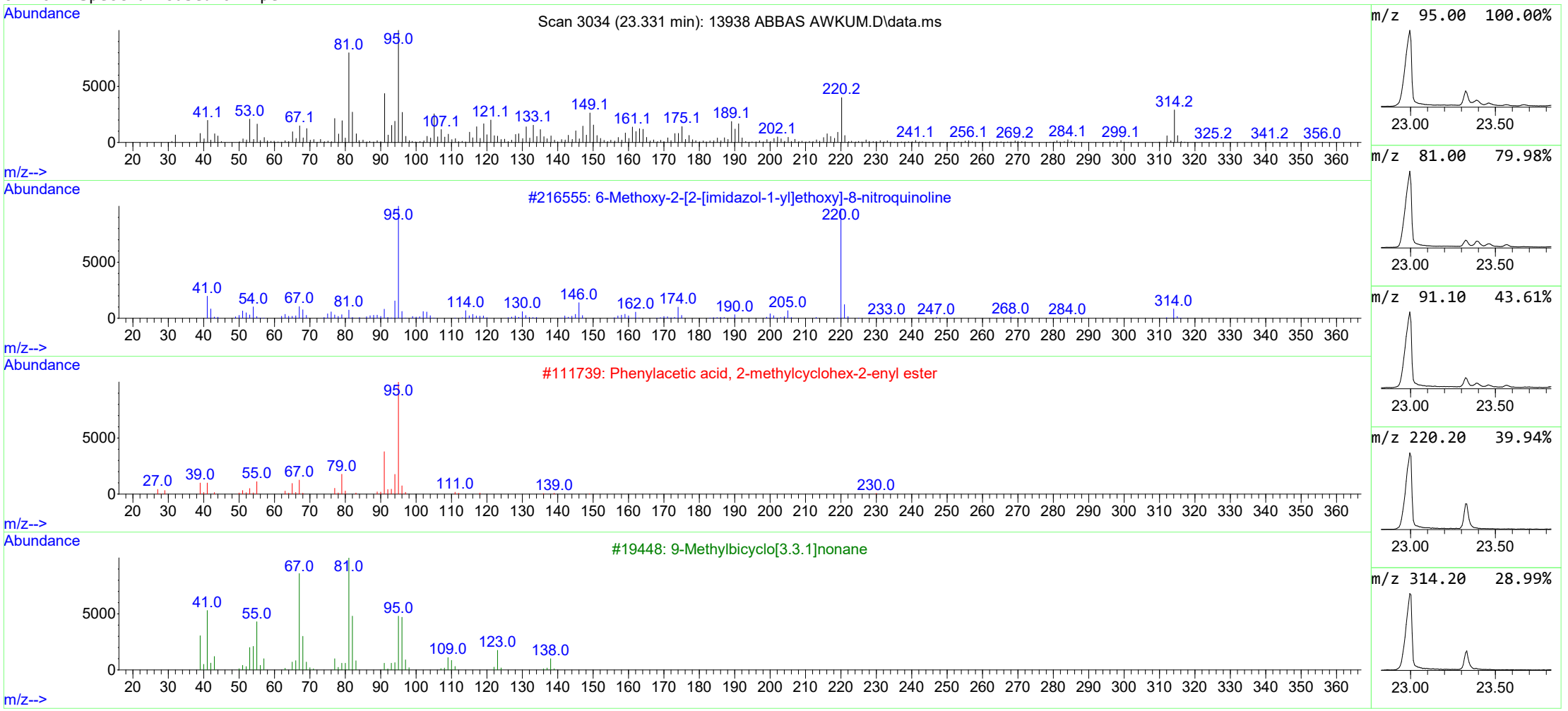


Data File: D:\GCMS RESULTS DATA\13938 ABBAS AWKUM.D
Sample : ABBAS

Peak Number: 9 at 22.998 min Area: 255333168 Area % 33.82

The 3 best hits from each library.	Ref\#	CAS\#	Qual
C:\MassHunter\LIBRARY\NIST20.L			
1 4-Cyclohexyl-1-(furan-2-ylmethyl...	198249	1000443-23-4	49
2 8-Phosphadispiro[2.1.2.2]nonane,...	94477	1010162-94-6	38
3 2H-1-Benzopyran-2-one, 4-methyl-...	94284	003993-57-5	25

Unknown Spectrum based on Apex

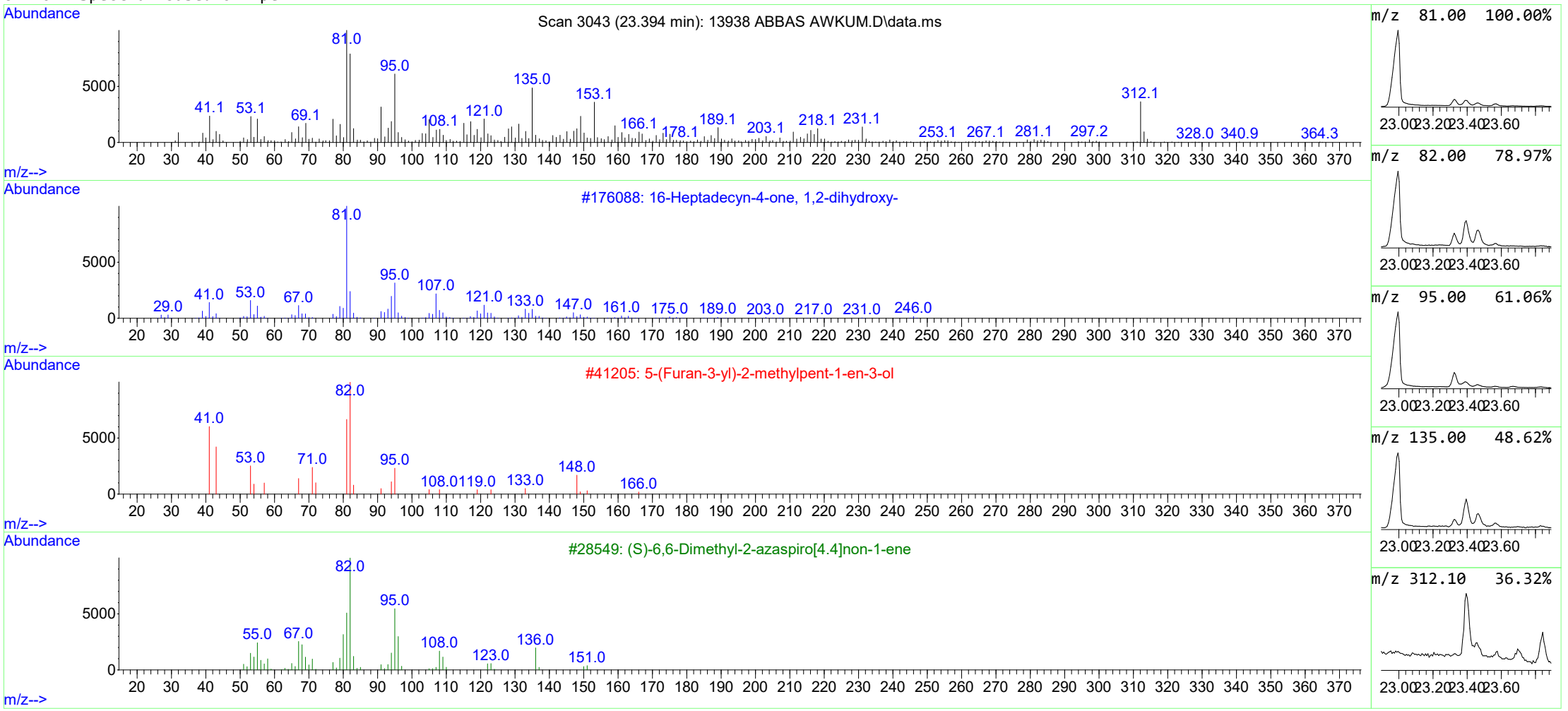


Data File: D:\GCMS RESULTS DATA\13938 ABBAS AWKUM.D
Sample : ABBAS

Peak Number: 10 at 23.329 min Area: 19057288 Area % 2.52

The 3 best hits from each library.	Ref\#	CAS\#	Qual
C:\MassHunter\LIBRARY\NIST20.L			
1 6-Methoxy-2-[2-[imidazol-1-yl]et...	216555	1000213-11-2	45
2 Phenylacetic acid, 2-methylcyclo...	111739	1000188-22-1	35
3 9-Methylbicyclo[3.3.1]nonane	19448	025107-01-1	30

Unknown Spectrum based on Apex

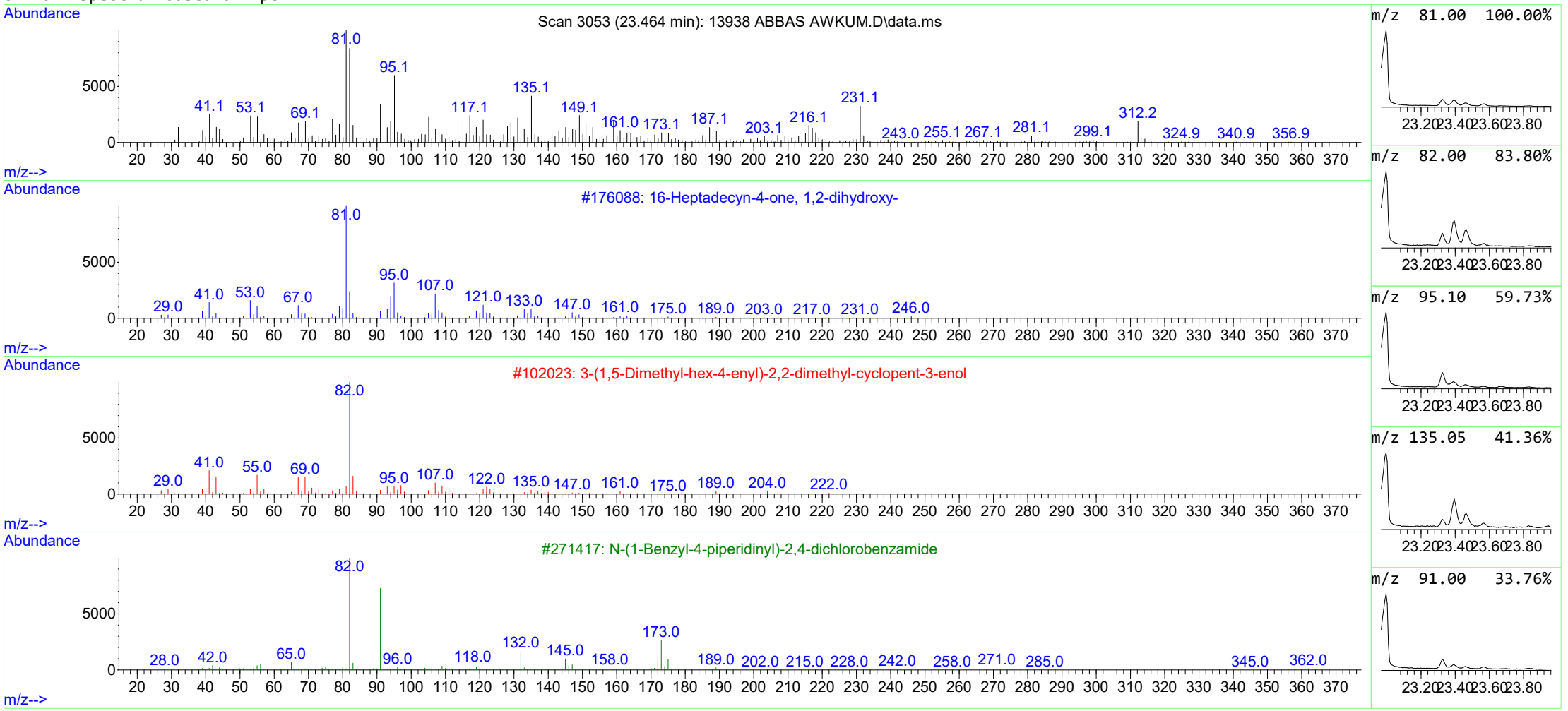


Data File: D:\GCMS RESULTS DATA\13938 ABBAS AWKUM.D
Sample : ABBAS

Peak Number: 11 at 23.394 min Area: 17938496 Area % 2.38

The 3 best hits from each library.	Ref\#	CAS\#	Qual
C:\MassHunter\LIBRARY\NIST20.L			
1 16-Heptadecyn-4-one, 1,2-dihydroxy-	176088	1379770-10-1	25
2 5-(Furan-3-yl)-2-methylpent-1-en...	41205	072075-28-6	14
3 (S)-6,6-Dimethyl-2-azaspiro[4.4]...	28549	055811-47-7	14

Unknown Spectrum based on Apex

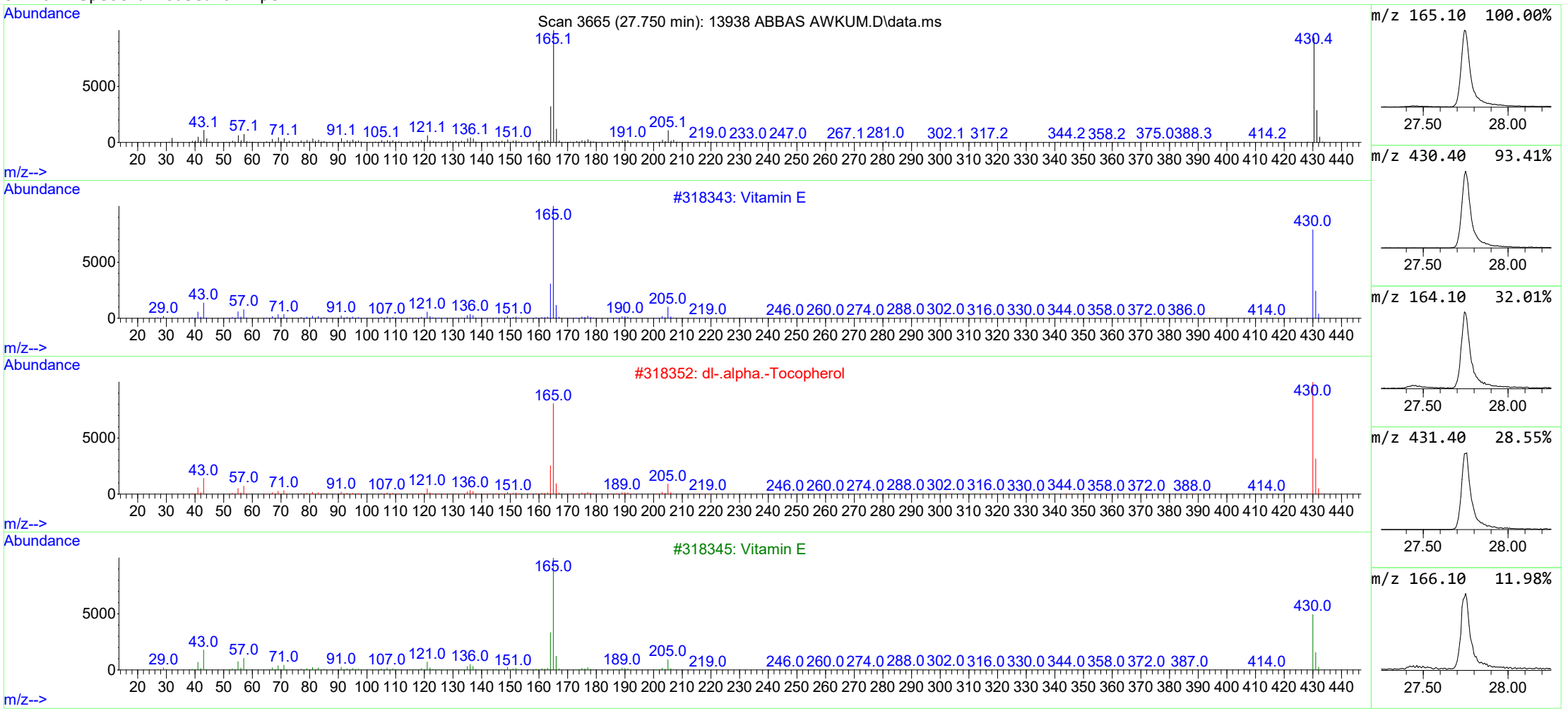


Data File: D:\GCMS RESULTS DATA\13938 ABBAS AWKUM.D
Sample : ABBAS

Peak Number: 12 at 23.464 min Area: 8623816 Area % 1.14

The 3 best hits from each library.	Ref\#	CAS\#	Qual
C:\MassHunter\LIBRARY\NIST20.L			
1 16-Heptadecyn-4-one, 1,2-dihydroxy-	176088	1379770-10-1	25
2 3-(1,5-Dimethyl-hex-4-enyl)-2,2-...	102023	1010189-03-6	25
3 N-(1-Benzyl-4-piperidinyl)-2,4-d...	271417	312281-74-6	15

Unknown Spectrum based on Apex



Data File: D:\GCMS RESULTS DATA\13938 ABBAS AWKUM.D
Sample : ABBAS

Peak Number: 13 at 27.750 min Area: 20342221 Area % 2.69

The 3 best hits from each library.	Ref\#	CAS\#	Qual
C:\MassHunter\LIBRARY\NIST20.L			
1 Vitamin E	318343	000059-02-9	99
2 dl-.alpha.-Tocopherol	318352	010191-41-0	99
3 Vitamin E	318345	000059-02-9	99

Centralized Resource Lab Library Search Report

Data Path : D:\GCMS RESULTS DATA\
Data File : 13938 ABBAS AWKUM.D
Acq On : 08 Sep 2026 00:12
Operator : CRL
Sample : ABBAS
Misc :
ALS Vial : 12 Sample Multiplier: 1

Search Libraries: C:\MassHunter\LIBRARY\NIST20.L Minimum Quality: 0

Unknown Spectrum: Apex
Integration Events: ChemStation Integrator - autoint1.e

Pk#	RT	Area%	Library/ID	Ref#	CAS#	Qual
1	15.614	14.12	C:\MassHunter\LIBRARY\NIST20.L			
			1-Butene, 1-(methylthio)-, (Z)-	4910	017414-15-2	47
			2-[2-[2-(2-Butoxyethoxy)ethoxy]eth	188598	1000351-93-9	43
			oxy]ethyl acetate			
			1,3-Dioxolane, 2-butyl-2-methyl-	24181	014447-27-9	43
2	18.066	0.92	C:\MassHunter\LIBRARY\NIST20.L			
			4-(2-Methyl-1H-imidazol-1-yl)anili	48224	1000484-51-3	58
			ne			
			Diallylphenylvinylsilane	92345	119901-89-2	50
			Phenylamine, 2-(5-methyl-2H-pyrazo	48241	1000309-95-3	43
			l-3-yl)-			
3	18.465	0.76	C:\MassHunter\LIBRARY\NIST20.L			
			Phytol	194548	000150-86-7	99
			Phytol	194550	000150-86-7	91
			Phytol	194552	000150-86-7	91
4	21.069	2.02	C:\MassHunter\LIBRARY\NIST20.L			
			5-Cyano-4-oxo-6-phenyl-2-thioxohex	112414	081585-29-7	25
			ahydropyrimidine			
			Furane-2-carboxaldehyde, 5-(2-meth	112532	329222-70-0	22
			yl-4-nitrophenyl)-			
			Phosphinic acid, diphenyl-, methyl	113667	001706-90-7	18
			ester			
5	21.989	10.39	C:\MassHunter\LIBRARY\NIST20.L			
			Bis(2-ethylhexyl) phthalate	295611	000117-81-7	91
			Bis(2-ethylhexyl) phthalate	295608	000117-81-7	91
			Phthalic acid, di(2-propylpentyl)	295658	1000377-93-5	91
			ester			
6	22.216	11.82	C:\MassHunter\LIBRARY\NIST20.L			
			3-Butenoic acid, 4-phenyl-, butyl	96620	054966-44-8	25
			ester			

Data Path : D:\GCMS RESULTS DATA\
Data File : 13938 ABBAS AWKUM.D
Acq On : 08 Sep 2026 00:12
Operator : CRL
Sample : ABBAS
Misc :
ALS Vial : 12 Sample Multiplier: 1

Search Libraries: C:\MassHunter\LIBRARY\NIST20.L Minimum Quality: 0

Unknown Spectrum: Apex
Integration Events: ChemStation Integrator - autoint1.e

Pk#	RT	Area%	Library/ID	Ref#	CAS#	Qual
			3-Heptanol, TMS derivative	62455	018132-91-7	25
			Quinoline, 6-methyl-, 1-oxide	35650	004053-42-3	14
7	22.344	11.66	C:\MassHunter\LIBRARY\NIST20.L			
			2-[(2-Furanylmethyl)amino]benzoic acid	95172	501661-50-3	30
			Benzoic acid, 3,5-bis(1,1-dimethyl ethyl)-2-hydroxy-	135795	019715-19-6	25
			1-Aminopyrene	95516	001606-67-3	25
8	22.485	5.77	C:\MassHunter\LIBRARY\NIST20.L			
			Tricyclo[3.2.1.0(2,4)]octane-3-carboxylic acid (furan-2-ylmethyl)-amide	112758	1000297-39-2	30
			Naphthalene, 1,2,3,4-tetrahydro-5-methoxy-	37870	001008-19-1	25
			(1R,3aS,5aS,8aR)-1,3a,4,5a-Tetramethyl-1,2,3,3a,5a,6,7,8-octahydrocyclopenta[c]pentalene	80449	065372-78-3	20
9	22.998	33.82	C:\MassHunter\LIBRARY\NIST20.L			
			4-Cyclohexyl-1-(furan-2-ylmethyl)-4H,5H,7H-pyrazolo[3,4-b]pyridin-6-one	198249	1000443-23-4	49
			8-Phosphadispiro[2.1.2.2]nonane, 8-phenyl-	94477	1010162-94-6	38
			2H-1-Benzopyran-2-one, 4-methyl-7-(2-propenyloxy)-	94284	003993-57-5	25
10	23.329	2.52	C:\MassHunter\LIBRARY\NIST20.L			
			6-Methoxy-2-[2-[imidazol-1-yl]ethoxy]-8-nitroquinoline	216555	1000213-11-2	45
			Phenylacetic acid, 2-methylcyclohex-2-enyl ester	111739	1000188-22-1	35
			9-Methylbicyclo[3.3.1]nonane	19448	025107-01-1	30

Data Path : D:\GCMS RESULTS DATA\
Data File : 13938 ABBAS AWKUM.D
Acq On : 08 Sep 2026 00:12
Operator : CRL
Sample : ABBAS
Misc :
ALS Vial : 12 Sample Multiplier: 1

Search Libraries: C:\MassHunter\LIBRARY\NIST20.L Minimum Quality: 0

Unknown Spectrum: Apex
Integration Events: ChemStation Integrator - autoint1.e

Pk#	RT	Area%	Library/ID	Ref#	CAS#	Qual
11	23.394	2.38	C:\MassHunter\LIBRARY\NIST20.L			
			16-Heptadecyn-4-one, 1,2-dihydroxy	176088	1379770-10-1	25
			5-(Furan-3-yl)-2-methylpent-1-en-3-ol	41205	072075-28-6	14
			(S)-6,6-Dimethyl-2-azaspiro[4.4]non-1-ene	28549	055811-47-7	14
12	23.464	1.14	C:\MassHunter\LIBRARY\NIST20.L			
			16-Heptadecyn-4-one, 1,2-dihydroxy	176088	1379770-10-1	25
			3-(1,5-Dimethyl-hex-4-enyl)-2,2-dimethyl-cyclopent-3-enol	102023	1010189-03-6	25
			N-(1-Benzyl-4-piperidinyl)-2,4-dichlorobenzamide	271417	312281-74-6	15
13	27.750	2.69	C:\MassHunter\LIBRARY\NIST20.L			
			Vitamin E	318343	000059-02-9	99
			dl-.alpha.-Tocopherol	318352	010191-41-0	99
			Vitamin E	318345	000059-02-9	99

TIC: 13938 ABBAS AWKUM.D\data.ms
ABBAS

Peak #	Ret Time	Type	Width	Area	Start Time	End Time
1	15.614	BV	0.332	106588174	14.974	15.797
2	18.066	BB	0.027	6965043	17.990	18.130
3	18.465	VV	0.037	5751144	18.423	18.550
4	21.069	BV	0.079	15242680	20.702	21.146
5	21.989	VV	0.030	78443017	21.921	22.114
6	22.216	PV	0.032	89233175	22.114	22.273
7	22.344	VV	0.029	88035449	22.273	22.440
8	22.485	VV	0.028	43531743	22.440	22.569
9	22.998	PV	0.043	255333168	22.842	23.164
10	23.329	VV	0.031	19057288	23.287	23.359
11	23.394	VV	0.040	17938496	23.359	23.434
12	23.464	VV	0.037	8623816	23.434	23.529
13	27.750	BB	0.057	20342221	27.673	27.930